

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Machine Learning

ASSESSMENT TOOL 1 – Analytical Problem Solving (5 QUESTIONS)

Course Code:	MLA0407	Course Title:	Deep Learning
CO Assessed:	CO1	Assessment:	Assessment Tool 1 – Analytical problem solving
Weightage:	35% of CIA	Total Marks:	(5 Questions × 10 Marks)
CO1 Statement:	<i>Apply the fundamental concepts of machine learning and neural networks to design and analyze learning algorithms using perceptrons, multilayer perceptrons, forward propagation, backpropagation, gradient descent optimization techniques, and Maximum Likelihood Estimation for solving real-world problems.</i>		
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Section / Batch:	2024	Date:	21/07/2026

Code of Conduct

I M. Kousik Reddy(192425185) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Kousik reddy

QUESTIONS: 5

Total Marks: 50

Annexure A

Analytical Problem Solving

Duration: 1hr

Q1.

Design a machine learning solution for predicting customer loan approval using a suitable learning algorithm. **Describe** the major stages involved in model development, including data preparation, model training, and performance evaluation. (10 marks)

Q2.

Describe the architecture of an Artificial Neural Network (ANN) consisting of input, hidden, and output layers. **Illustrate** the role of artificial neurons, weights, bias, and activation functions in processing input data to generate predictions. (10 marks)

Q3.

Analyze the working of the Perceptron learning algorithm for solving a binary classification problem. **Demonstrate** the weight and bias update process for one iteration using a suitable numerical example. (10 marks)

Q4.

Illustrate the forward propagation process in a Multilayer Perceptron (MLP) using a simple neural network with one hidden layer. **Compute** the output values for the given inputs, weights, biases, and activation function. (10 marks)

Q5.

Demonstrate the Backpropagation algorithm for training a neural network by calculating the output error, propagating the error through hidden layers, and updating the network weights using Gradient Descent. (10 marks)

Q6.

Compare and evaluate Batch Gradient Descent, Stochastic Gradient Descent (SGD), and Mini-Batch Gradient Descent with respect to convergence behavior, computational efficiency, memory requirements, and practical applications in deep learning. (10 marks)

Q7.

Formulate a Maximum Likelihood Estimation (MLE) approach for estimating the parameters of a probability distribution from a given dataset. **Interpret** the estimated parameters and discuss their significance in machine learning model development. (10 marks)

Q8.

Analyze the impact of the Curse of Dimensionality on machine learning models with respect to training complexity, prediction accuracy, and data sparsity. **Propose** suitable techniques to overcome these challenges in real-world applications. (10 marks)

Q9.

Design a Multilayer Perceptron (MLP) model for a multiclass image classification problem.

Describe the network architecture and explain the role of forward propagation, backpropagation, and Gradient Descent during the training process. (10 marks)

Q10.

Evaluate the importance of learning algorithms in building intelligent machine learning systems. **Compare** traditional machine learning approaches with neural network-based learning by considering model complexity, learning capability, and application scenarios. (10 marks)

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Technical Content Accuracy	30	Highly accurate, indepth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, wellanalyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides

Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness
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Faculty In-charge	Course Coordinator	Program Director
Signature: _____ Date: _____	Signature: _____ Date: _____	Signature: _____ Date: _____

Q1. Design a Machine Learning Solution for Customer Loan Approval (10 Marks)

Introduction

Loan approval prediction is a **supervised machine learning classification problem** where the system predicts whether a customer's loan should be **Approved** or **Rejected** based on customer information.

Suitable Algorithm

- Decision Tree
- Random Forest
- Logistic Regression
- Artificial Neural Network (ANN)

(Random Forest is commonly used because it provides high accuracy.)

Stages of Model Development

1. Data Collection

Collect customer information:

- Age
- Income
- Employment Status
- Credit Score
- Loan Amount
- Previous Loan History

Target:

- Loan Approved (Yes/No)

2. Data Preparation

- Remove duplicate records
- Handle missing values
- Encode categorical values
- Normalize numerical values
- Split data:
 - Training = 80%
 - Testing = 20%

3. Model Training

Train the algorithm using the training dataset.

Customer Data



Machine Learning Algorithm



Trained Model

4. Prediction

Example:

Income = ₹50,000

Credit Score = 760

Employment = Permanent

Prediction:

Loan Approved = YES

5. Performance Evaluation

Metrics:

- Accuracy
- Precision
- Recall
- F1 Score
- Confusion Matrix

Accuracy Formula

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

Example:

Accuracy = 96%

Advantages

- Fast loan approval
- Reduces human effort
- High prediction accuracy
- Consistent decisions

Conclusion

Machine learning automates loan approval by learning patterns from historical customer data, making the approval process faster and more accurate.

Q2. Architecture of Artificial Neural Network (ANN) (10 Marks)

Introduction

Artificial Neural Network (ANN) is a computational model inspired by the human brain. It consists of interconnected neurons arranged in layers.

Architecture

Input Layer

X1 X2 X3

\\ /

Hidden Layer

H1 H2

\\ /

Output Layer

Y

Components

1. Input Layer

Receives input features.

Example:

- Age
- Salary
- Credit Score

2. Hidden Layer

Processes input data and extracts useful features.

3. Output Layer

Produces the final prediction.

Example:

- Loan Approved
- Loan Rejected

Artificial Neuron

Each neuron calculates

$$Z = \sum W_i X_i + b$$

Where

- X = Input
- W = Weight
- b = Bias

Weights

Weights indicate the importance of each input feature.

Bias

Bias shifts the activation function and improves learning.

Activation Functions

Sigmoid

$$\sigma(x) = \frac{1}{1 + e^{-x}}$$

Range: 0–1

ReLU

$$ReLU(x) = \max(0, x)$$

Tanh

Range: -1 to 1

Example

Given:

- $X_1 = 2$
- $X_2 = 3$
- $W_1 = 0.4$
- $W_2 = 0.6$
- Bias = 1

Net Input:

$$\begin{aligned} Z &= (2 \times 0.4) + (3 \times 0.6) + 1 \\ &= 0.8 + 1.8 + 1 = 3.6 \end{aligned}$$

Sigmoid Output:

$$\sigma(3.6) = 0.973$$

Prediction = Positive

Advantages

- Learns complex patterns
- High prediction accuracy

- Suitable for image and speech recognition

Conclusion

ANN uses neurons, weights, bias, and activation functions to learn complex relationships and generate predictions.

Q3. Perceptron Learning Algorithm (10 Marks)

Introduction

The Perceptron is the simplest neural network used for binary classification.

Outputs:

- 0
- 1

Perceptron Structure

Inputs

X1 X2

\ /

Weights

↓

Summation

↓

Activation

↓

Output

Algorithm Steps

Step 1: Initialize

$W_1 = 0.2$

$W_2 = 0.3$

$\text{Bias} = 0.1$

$\text{Learning Rate} = 0.1$

Step 2: Input

$X_1 = 1$

$X_2 = 0$

$\text{Target} = 1$

Step 3: Net Input

$$\begin{aligned} Z &= W_1X_1 + W_2X_2 + b \\ &= (0.2)(1) + (0.3)(0) + 0.1 = 0.3 \end{aligned}$$

Step 4: Activation

Since

$$Z > 0$$

Output = 1

Step 5: Error

$$\begin{aligned} \text{Error} &= \text{Target} - \text{Output} \\ &= 1 - 1 = 0 \end{aligned}$$

No weight update required.

Weight Update Example

Suppose

Target = 1

Predicted = 0

Error = 1

Weight Update Formula

$$W_{new} = W_{old} + \eta \times Error \times Input$$

Update W_1

$$= 0.2 + 0.1(1)(1) = 0.3$$

Update W_2

$$= 0.3 + 0.1(1)(0) = 0.3$$

Bias Update

$$b_{new} = 0.1 + 0.1 = 0.2$$

Updated Values

$W_1 = 0.3$

$W_2 = 0.3$

Bias = 0.2

Advantages

- Simple algorithm
- Fast learning
- Easy implementation

Limitations

- Solves only linearly separable problems
- Cannot solve XOR problem

Conclusion

The perceptron updates weights and bias whenever an incorrect prediction occurs, enabling it to learn binary classification tasks.

Q4. Forward Propagation in Multilayer Perceptron (10 Marks)

Introduction

Forward propagation is the process of passing input data through the network to obtain the output.

Network

Input $\rightarrow$ Hidden Layer $\rightarrow$ Output

Given:

- Inputs: $X_1 = 1, X_2 = 2$
- Hidden Weights:
 - $W_1 = 0.2$
 - $W_2 = 0.3$

- Bias = 0.1
- Activation = Sigmoid

Hidden Layer

$$Z_h = (1 \times 0.2) + (2 \times 0.3) + 0.1 = 0.9$$

Sigmoid:

$$H = \frac{1}{1 + e^{-0.9}} = 0.711$$

Output Layer

Weight = 0.5

Bias = 0.2

$$Z_o = (0.711 \times 0.5) + 0.2 = 0.5555$$

Output

$$Y = \frac{1}{1 + e^{-0.5555}} = 0.635$$

Conclusion

Forward propagation computes outputs layer by layer using weights, bias, and activation functions.

Q5. Backpropagation Algorithm (10 Marks)

Steps

1. Forward Propagation

Calculate network output.

Suppose

Target = 1

Output = 0.8

2. Output Error

$$\begin{aligned} \text{Error} &= \text{Target} - \text{Output} \\ &= 1 - 0.8 = 0.2 \end{aligned}$$

3. Compute Gradient

$$\begin{aligned} \text{Gradient} &= \text{Error} \times \text{Output}(1 - \text{Output}) \\ &= 0.2 \times 0.8 \times 0.2 = 0.032 \end{aligned}$$

4. Weight Update

Learning Rate = 0.1

Old Weight = 0.5

Input = 0.7

$$\begin{aligned} W_{\text{new}} &= 0.5 + 0.1 \times 0.032 \times 0.7 \\ &= 0.50224 \end{aligned}$$

5. Repeat

Continue until error becomes minimum.

Advantages

- Reduces prediction error
- Improves model accuracy
- Learns complex patterns

Conclusion

Backpropagation updates weights using gradients to minimize the loss function.

Q6. Comparison of Gradient Descent Methods (10 Marks)

Feature	Batch GD	SGD	Mini-Batch GD
Data Used	Entire dataset	One sample	Small batch
Speed	Slow	Fast	Moderate
Memory	High	Low	Medium
Convergence	Stable	Noisy	Stable
Accuracy	High	Moderate	High
Applications	Small datasets	Online learning	Deep Learning

Conclusion

Mini-Batch Gradient Descent is the most widely used optimization method because it balances speed, memory efficiency, and convergence.

Q7. Maximum Likelihood Estimation (MLE) (10 Marks)

Definition

MLE estimates model parameters that maximize the likelihood of observing the given data.

Example Dataset

5, 6, 7, 8, 9

Mean Estimate

$$\mu = \frac{5 + 6 + 7 + 8 + 9}{5} = 7$$

Variance

$$\begin{aligned}\sigma^2 &= \frac{\sum (x - \mu)^2}{n} \\ &= \frac{10}{5} = 2\end{aligned}$$

Estimated Parameters

- Mean = 7
- Variance = 2

Importance

- Parameter estimation
- Model fitting
- Used in Logistic Regression, Naive Bayes, Gaussian models

Conclusion

MLE finds the parameter values that best explain the observed data.

Q8. Curse of Dimensionality (10 Marks)

Definition

As the number of features increases, machine learning becomes more difficult due to sparse data and increased computation.

Effects

- Increased training time
- High memory usage
- Lower prediction accuracy
- Data sparsity
- Overfitting

Solutions

- Feature Selection
- Principal Component Analysis (PCA)
- Regularization
- Collect more data
- Dimensionality Reduction

Applications

- Image Recognition
- Text Mining
- Bioinformatics

Conclusion

Reducing unnecessary features improves efficiency and model performance.

Q9. MLP for Multiclass Image Classification (10 Marks)

Architecture

Image Pixels



Input Layer



Hidden Layer 1



Hidden Layer 2



Output Layer

(Cat Dog Car)

Training Process

Forward Propagation

- Computes predictions.

Loss Function

- Calculates prediction error.

Backpropagation

- Computes gradients.

Gradient Descent

- Updates weights to reduce error.

Activation Functions

- ReLU (Hidden Layers)
- Softmax (Output Layer)

Applications

- Face Recognition
- Traffic Sign Detection
- Medical Image Classification

Conclusion

MLPs classify images into multiple classes by repeatedly updating weights during training.

Q10. Importance of Learning Algorithms (10 Marks)

Learning Algorithms

Learning algorithms enable machines to identify patterns, make predictions, and improve performance from data.

Comparison

Feature	Traditional ML	Neural Networks
Complexity	Low	High
Feature Engineering	Manual	Automatic
Data Requirement	Small	Large
Training Time	Fast	Slow
Accuracy	Moderate	High
Best Applications	Structured Data	Images, Speech, NLP

Importance

- Automation
- High prediction accuracy
- Handles large datasets
- Learns complex relationships
- Supports intelligent systems

Conclusion

Traditional machine learning is suitable for smaller structured datasets, while neural network-based learning excels in solving complex problems such as image recognition, speech processing, and natural language processing.

